

Notes: Hayashi (Econometrica, 1982)

Tobin's Marginal q and Average q : A Neoclassical Interpretation

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1 Big picture

- **Question.** Can Tobin's q -theory of investment be derived cleanly from a firm's dynamic optimization problem, and when is the *observable* average q a valid measure of the *unobservable* marginal q that actually governs investment?
- **Why this matters.** Empirical work often regresses investment on average q , because marginal q cannot be observed directly. Hayashi asks when that practice is theoretically justified.
- **Setting.** The paper builds a dynamic investment model with installation costs, corporate taxes, depreciation allowances, and investment tax credits. It then adds a short empirical illustration using annual data for the U.S. corporate sector from 1952–1978.
- **Core challenge.** Tobin's original discussion says investment depends on the market value of an additional unit of capital relative to replacement cost. But firms and researchers usually observe only the total market value of the firm relative to replacement cost of the existing capital stock.
- **Main conceptual contribution.** Hayashi shows that the modern neoclassical model with adjustment costs and Tobin's q -theory are really the same object. The firm's optimal investment rule can be written as a function of *modified marginal q* , where the modification adjusts for tax rules.
- **Main theoretical result on investment.** Under a very general installation-cost model,

$$I = \alpha(\tilde{q}, K; t), \quad \tilde{q} \equiv \frac{q}{1 - k - z}.$$

Here q is marginal q , k is the investment tax credit rate, and z is the present value of tax deductions attached to one additional unit of investment.

- **Main theoretical result on average versus marginal q .** If the firm is a price taker and both the production function and the installation function are linearly homogeneous, then marginal q equals average q *up to a tax term*. Without taxes, the famous equality $q = h$ holds exactly.
- **Monopoly case.** If the firm is a price maker, then average q is strictly above marginal q by a monopoly-rent component. So average q is no longer a sufficient statistic for investment opportunities.
- **Tax adjustment is central.** Even in the competitive constant-returns case, raw average q is not the right investment signal. The correct object subtracts the capitalized value of tax deductions on past investment and divides by $(1 - k - z)$.
- **Empirical implementation.** Hayashi computes modified q for the U.S. corporate sector using the postwar U.S. tax code, depreciation rules, and an existing series for average q .
- **Main empirical illustration.** Using annual aggregate data, Hayashi estimates

$$I/K = 0.0980 + 0.0423 \tilde{q},$$

with $R^2 = 0.46$ and a very low Durbin–Watson statistic of 0.43.

- **Interpretation of the empirical result.** Modified q has economically meaningful explanatory power for aggregate investment, but the aggregate regression is only a rough check rather than a

full structural validation.

- **Economic takeaway.** The paper gives the classic microfoundation for investment regressions based on q , but it also disciplines them sharply: average q can stand in for marginal q only under restrictive technological and market-structure assumptions, and tax adjustments are not optional.

2 Data and measurement (key objects)

2.1 This is mainly a theory paper

Hayashi’s paper is not built around a firm-level micro dataset or a modern identification design. The main “objects” are model primitives and tax-adjusted valuation concepts. The empirical material is an aggregate annual time series for the U.S. corporate sector.

2.2 Firm-side state and control variables

- $K(t)$: capital stock.
- $I(t)$: gross investment.
- $N(t)$: vector of variable inputs, such as labor.
- $p(t)$: output price.
- $p_I(t)$: price of investment goods.
- $r(t)$: nominal discount rate.
- δ : physical depreciation rate.

Interpretation. The capital stock is the slow-moving state variable; investment is the control margin. Once adjustment costs are introduced, the firm chooses how fast to add capital rather than jumping directly to a frictionless target stock.

2.3 Production and profit

Before tax, operating profits are

$$\pi(t) = p(t)F(K(t), N(t), t) - w(t)N(t), \quad (1)$$

where F is the production function and $w(t)$ is the vector of input prices.

Interpretation. The model is flexible enough to allow either a competitive firm or a price-making firm. That distinction matters later for the relation between marginal and average q .

2.4 Tax variables

- $u(t)$: corporate income tax rate.

- $k(t)$: investment tax credit rate.
- $D(x, t - x)$: depreciation allowance for tax purposes on an asset of age x , under the tax code in force when the asset was purchased.

Interpretation. Taxes affect the shadow value of capital. The investment-relevant q is therefore not simply a raw valuation ratio; it must be adjusted for tax shields and credits.

2.5 Two ways to represent adjustment costs

Hayashi discusses two equivalent styles:

- **Convex installation cost in profits:** $G(I, K; t)$ enters the profit function directly.
- **Installation function in capital accumulation:** only $\psi(I, K; t)$ units of effective capital are created from I units of gross investment.

Interpretation. Hayashi works mainly with the second representation because it makes the connection to marginal q especially transparent.

2.6 The installation function

Under Uzawa's formulation,

$$\dot{K} = \psi(I, K; t) - \delta K. \quad (2)$$

The derivative ψ_I is increasing but concave in I , and it can fall sharply when I becomes negative, reflecting irreversibility.

Interpretation. Gross investment is not transformed into productive capital one-for-one. At high investment rates, installation becomes less efficient at the margin.

2.7 Marginal q , average q , and modified q

Hayashi defines:

$$q \equiv \frac{\lambda}{p_I}, \quad (3)$$

where λ is the shadow value of installed capital, and

$$h \equiv \frac{V}{p_I K}, \quad (4)$$

where V is firm value.

The tax-adjusted object governing investment is

$$\tilde{q} \equiv \frac{q}{1 - k - z}. \quad (5)$$

Interpretation.

- q : market value of one extra unit of installed capital relative to replacement cost.

- h : total market value of the firm relative to replacement cost of its existing capital stock.
- $\tilde{q}$: marginal q adjusted for taxes and credits.

2.8 The tax-shield objects $A(0)$ and $z(t)$

- $A(0)$: the present discounted value of current and future tax deductions attributable to *past* investments.
- $z(t)$: the present discounted value of tax deductions generated by *one additional unit of current investment*.

Interpretation. $A(0)$ matters for the mapping from average q to marginal q , because market value capitalizes old tax shields. $z(t)$ matters for the current investment decision, because it lowers the effective cost of new capital.

2.9 Aggregate empirical series in Section 4

The empirical illustration uses annual U.S. corporate-sector data for 1952–1978. Hayashi constructs or imports:

- A_t : capitalized tax deductions from past investment,
- $p_I K$: replacement cost of capital,
- $a_t \equiv A_t / (p_I K)$,
- z_t : tax value of depreciation allowances on new investment,
- k_t : effective investment tax credit rate,
- h_t : average q from von Furstenberg,
- $\tilde{q}_t$: modified q ,
- $p_I I_t$: investment spending.

2.10 Assumptions used to build the tax-adjusted series

- Future corporate tax rate is assumed to remain at 48%.
- The discount rate used in the tax calculations is the Baa corporate bond rate plus 4 percentage points.
- Straight-line depreciation is assumed before 1954.
- Sum-of-years-digits depreciation is assumed after 1954.
- Asset-specific data are aggregated across producers' durable equipment, nonresidential structures, and residential structures.

Interpretation. The empirical section is best read as a tax-accounting adjustment exercise that turns observable average q into an approximation of the theoretically relevant $\tilde{q}$.

3 Models and empirical specifications (all equations + interpretation)

3.1 Firm value maximization

The firm chooses investment and variable inputs to maximize the present discounted value of after-tax net receipts:

$$V(0) = \int_0^\infty R(t) \exp\left(-\int_0^t r(s) ds\right) dt. \quad (6)$$

Interpretation. This is a standard continuous-time present-value problem. All of the paper's main results are obtained from this dynamic optimization setup.

3.2 Net receipts with taxes and tax credits

Net receipts are

$$R(t) = [1 - u(t)]\pi(t) + u(t) \int_0^\infty D(x, t - x) p_I(t - x) I(t - x) dx - [1 - k(t)]p_I(t)I(t). \quad (7)$$

Interpretation.

- The first term is after-tax operating profit.
- The second term adds depreciation deductions on past investment.
- The third term subtracts current investment spending net of the investment tax credit.

3.3 Baseline profit and the old capital-accumulation law

Operating profit is

$$\pi(t) = p(t)F(K(t), N(t), t) - w(t)N(t), \quad (8)$$

and the frictionless accumulation law is

$$\dot{K} = I - \delta K. \quad (9)$$

Interpretation. Under this older formulation the optimal *stock* of capital may be pinned down, but the optimal *rate* of investment is not. This is Hayashi's motivation for bringing adjustment costs inside the optimization problem.

3.4 Adjustment costs via convex installation costs

One possible representation is

$$\pi = p[F(K, N; t) - G(I, K; t)] - wN, \quad (10)$$

where $G_I > 0$ and $G_{II} > 0$.

Interpretation. The faster the firm tries to invest, the more costly installation becomes at the margin.

3.5 Adjustment costs via the installation function

Hayashi then switches to Uzawa's formulation:

$$\dot{K} = \psi(I, K; t) - \delta K. \quad (11)$$

Interpretation. This is the formulation used for the main results. Effective capital created from investment is $\psi(I, K; t)$, not simply I . The derivative ψ_I is the efficiency with which one extra unit of gross investment is turned into installed capital.

3.6 Reduced objective and the tax value of new investment

After rearranging the present-value problem, Hayashi writes it as

$$V(0) = A(0) + \int_0^\infty \left[(1-u)\pi - (1-k-z)p_I I \right] \exp\left(-\int_0^t r(s) ds\right) dt, \quad (12)$$

where

$$z(t) = \int_0^\infty u(t+x)D(x, t) \exp\left[-\int_0^x r(t+s) ds\right] dx. \quad (13)$$

Interpretation.

- $A(0)$ is a sunk-value term tied to old investment and old tax deductions.
- $z(t)$ is the present value of the depreciation deductions created by one new unit of investment.
- The effective purchase price of investment goods from the firm's point of view is $(1-k-z)p_I$, not simply p_I .

3.7 First-order conditions

Under the installation-function formulation, the optimality conditions are:

$$\pi_N = 0, \quad (14)$$

$$(1-k-z)p_I = \lambda \psi_I, \quad (15)$$

$$\dot{\lambda} = (r + \delta - \psi_K)\lambda - (1-u)\pi_K, \quad (16)$$

with transversality condition

$$\lim_{t \rightarrow \infty} \lambda(t) K(t) \exp\left(-\int_0^t r(s) ds\right) = 0. \quad (17)$$

Interpretation.

- $\pi_N = 0$ is the usual static first-order condition for variable inputs.

- The investment condition equates the tax-adjusted cost of one more unit of investment goods to the shadow value of the installed capital it creates.
- The costate equation says the shadow value of capital depends on discounting, physical depreciation, the effect of capital on installation efficiency, and the extra profits generated by more capital.

3.8 Defining marginal q , average q , and modified q

Hayashi defines Tobin's marginal q as

$$q = \frac{\lambda}{p_I}, \quad (18)$$

and average q as

$$h = \frac{V}{p_I K}. \quad (19)$$

From the investment first-order condition,

$$\frac{q}{1 - k - z} = \frac{1}{\psi_I} \equiv \tilde{q}. \quad (20)$$

Interpretation. Modified q is the tax-adjusted version of marginal q . It tells the firm how valuable one more installed unit of capital is relative to the tax-adjusted acquisition cost of new capital.

3.9 The optimal investment rule

Because ψ_I can be inverted, the firm's optimal investment rule can be written as

$$I = \alpha(\tilde{q}, K; t). \quad (21)$$

If the installation function is linearly homogeneous in I and K , this becomes

$$\frac{I}{K} = \beta(\tilde{q}; t). \quad (22)$$

Interpretation. This is one of the deepest results in the paper. Once modified q is known, all the information from technology, demand, and expectations that matters for current investment is summarized in that single state variable.

3.10 Proposition 1: when marginal q and average q coincide

If the firm is a price taker, the transversality condition holds, and both the production and installation functions are linearly homogeneous, then

$$q(0) = h(0) - \frac{A(0)}{p_I(0)K(0)}. \quad (23)$$

Writing

$$a(0) \equiv \frac{A(0)}{p_I(0)K(0)}, \quad (24)$$

this becomes

$$q(0) = h(0) - a(0). \quad (25)$$

Interpretation.

- With taxes, raw average q overstates the relevant marginal q by the value of old tax shields.
- Without taxes and depreciation adjustments, $A(0) = 0$, so the famous identity $q = h$ emerges exactly.
- The result depends critically on competition and constant returns in both production and installation.

3.11 Corollary: the empirical investment equation under the competitive benchmark

Under the Proposition 1 conditions,

$$\tilde{q} = \frac{h - a}{1 - k - z}, \quad (26)$$

so the investment rule becomes

$$\frac{I}{K} = \beta \left(\frac{h - a}{1 - k - z}; t \right). \quad (27)$$

Interpretation. This is the key bridge from theory to empirics. It says observed average q can be used only after subtracting the capitalization of old tax deductions and correcting for current tax incentives.

3.12 Proposition 2: the price-making case

If the firm is a price maker, while production and installation remain linearly homogeneous, then Hayashi shows

$$q(0) = h(0) - a(0) - \frac{1}{p_I(0)K(0)} \int_0^\infty \eta(t)(1 - u(t))p(t)F(t) \exp\left(-\int_0^t r(s) ds\right) dt, \quad (28)$$

where

$$\eta(t) \equiv -\frac{F}{p} \frac{dp}{dF} \quad (29)$$

is the inverse elasticity term from demand.

Interpretation. The extra integral is monopoly rent. Therefore average q is too high relative to marginal q , because the total market value of the firm contains inframarginal monopoly rents that do not measure the value of one more installed unit of capital.

3.13 Empirical implementation

Hayashi computes tax-adjusted series for the U.S. corporate sector and then estimates a simple linear investment equation:

$$\frac{I}{K} = 0.0980 + 0.0423 \tilde{q}, \quad (30)$$

with standard errors (0.00840) and (0.00912), sample period 1953–1976, $R^2 = 0.46$, and Durbin–Watson statistic 0.43.

Interpretation. The regression is not the main contribution. It simply shows that the theory-driven modified q is empirically nontrivial and tracks aggregate investment in a meaningful way, though serial correlation remains severe.

4 Identification and instruments (what makes the estimates credible)

4.1 This is not an IV paper

Hayashi does not identify a causal effect using instruments, natural experiments, or panel variation. The paper’s credibility is primarily **theorem-based**. The strongest results are logical consequences of the dynamic optimization problem plus a small set of economically transparent assumptions.

4.2 What identifies the investment rule

The optimal investment rule

$$I = \alpha(\tilde{q}, K; t)$$

is identified by the first-order condition

$$(1 - k - z)p_I = \lambda\psi_I.$$

Given the installation function, ψ_I maps the shadow value of capital into the investment rate.

Why this is powerful. The paper does *not* need a special functional form for the production function or the demand curve in order to show that investment is a function of modified q . Those deeper primitives affect q itself, but not the form of the policy rule once q is known.

4.3 What identifies the equality between average and marginal q

The result $q = h - a$ rests on three knife-edge conditions:

- the firm is a price taker,
- the production function is linearly homogeneous,
- the installation function is linearly homogeneous.

Interpretation. These conditions guarantee that scaling the firm up or down proportionally does

not change average firm value per unit of capital. Once that scaling property holds, the market value of an additional unit of capital equals the average market value of existing capital, after correcting for tax shields.

4.4 Why monopoly breaks the average- q proxy

If the firm is a price maker, an expansion in output lowers the output price. Then the value of one more unit of capital is smaller than the average value of the old capital stock. Average q now embeds monopoly rents, so it is no longer a clean marginal signal.

Interpretation. This is the paper's main warning to empirical users of average q : even if average q is easy to observe, it is conceptually wrong whenever the firm's valuation includes rents that are not marginally reproducible.

4.5 Why tax adjustments are essential rather than cosmetic

Even in the competitive constant-returns benchmark, firm value includes the present value of depreciation deductions attached to old investment. Hayashi strips those out using $A(0)$, and he adjusts the effective acquisition cost of new capital using k and z .

Interpretation. The relevant empirical variable is not raw average q , but tax-adjusted modified q . If one skips these corrections, the theory says the regression variable is misspecified.

4.6 What makes the empirical illustration credible

- It uses the actual postwar U.S. corporate tax system, not a hypothetical flat tax with no depreciation rules.
- It separately calculates tax quantities across major asset classes and then aggregates them.
- It transparently states the expectation assumptions needed to discount future tax deductions.

4.7 Main limitations

- The empirical section is aggregate, so it cannot distinguish across industries or firms.
- The regression is reduced-form and very simple.
- The Durbin–Watson statistic of 0.43 indicates strong serial correlation.
- The tax computations assume static expectations for the discount rate and a constant future corporate tax rate of 48%.
- The bridge from average q to marginal q still depends on the strong benchmark of competition and constant returns.

Bottom line. The paper is most credible as a foundational theory paper. The empirical illustration is supportive, but the real force of the paper lies in clarifying exactly when investment regressions on average q do and do not make sense.

5 Tables and figures

5.1 Figure 1. The installation function

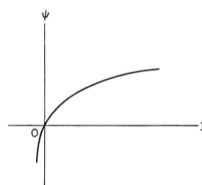


FIGURE 1.

Figure 1: Hayashi's schematic installation function ψ_I . The curve is increasing and concave in investment, capturing adjustment costs and partial irreversibility.

Read.

- The horizontal axis is gross investment I , and the vertical axis is the marginal conversion of gross investment into effective capital.
- The curve is concave for positive investment, so the marginal efficiency of installation falls as investment rises.
- The curve drops sharply as investment becomes negative, reflecting irreversibility or the high cost of disinvestment.

Economic message. This picture is the whole intuition behind q -theory. The firm does not buy capital one-for-one; it buys gross investment goods and then pays installation frictions. Marginal q measures the value of relaxing that margin.

5.2 Table I. Tax adjustments and modified q , 1952–1978

Read.

- The ratio $a_t = A_t/(p_I K)$ rises from about 0.21 in 1952 to around 0.29 in the early 1960s, then falls to about 0.13 by 1978.
- The present value of new-investment tax deductions z_t stays economically large, roughly in the 0.30–0.35 range for most of the sample.
- The effective investment tax credit k_t is zero before 1962, then becomes positive and rises to roughly 7% by the late 1970s.
- Average q and modified q both rise into the mid-1960s and then decline, but modified q moves less dramatically than raw average q .
- Modified q peaks around the mid-1960s, reaching about 1.36 in 1966 in Hayashi's table.

Economic message. Taxes materially change the mapping from stock-market valuation to investment incentives. Inflation and depreciation rules alter the size of tax shields, while investment tax credits change the effective purchase price of new capital. That is why modified q is smoother and conceptually better behaved than raw average q .

TABLE I

Year	(1) A	(2) $p_t K$	(3) a	(4) z	(5) k	(6) h	(7) $\tilde{q}$	(8) $p_t I$
1952	36.74	172.27	.213	.298	.0	.564	.499	22.58
1953	36.75	182.63	.201	.294	.0	.596	.559	24.88
1954	44.35	191.57	.232	.309	.0	.587	.515	24.59
1955	49.74	202.25	.246	.316	.0	.711	.680	27.75
1956	54.60	227.18	.240	.318	.0	.774	.783	32.46
1957	58.81	250.49	.235	.314	.0	.727	.718	34.78
1958	65.52	261.59	.250	.321	.0	.674	.624	30.09
1959	69.25	270.36	.256	.330	.0	.807	.822	32.66
1960	73.86	278.43	.265	.329	.0	.842	.859	35.22
1961	79.16	286.35	.276	.329	.0	.853	.859	35.04
1962	83.30	295.30	.282	.330	.0216	.954	1.036	38.68
1963	88.01	307.90	.288	.332	.0276	.898	.956	40.58
1964	91.98	322.59	.285	.354	.0289	1.009	1.174	45.58
1965	96.87	342.97	.282	.354	.0314	1.062	1.268	54.71
1966	100.70	376.77	.267	.345	.0319	1.116	1.363	62.94
1967	106.27	418.58	.254	.339	.0325	.919	1.058	63.79
1968	109.82	461.67	.238	.332	.0341	.949	1.121	69.91
1969	113.10	514.89	.220	.322	.0244	.979	1.162	78.23
1970	115.85	577.54	.201	.327	.0110	.789	.888	78.75
1971	123.28	637.27	.193	.332	.0196	.761	.875	83.34
1972	129.83	689.65	.188	.336	.0321	.794	.959	94.15
1973	136.33	756.54	.180	.336	.0339	.869	1.094	110.72
1974	141.95	903.56	.157	.327	.0347	.753	.933	121.64
1975	149.12	1074.71	.139	.317	.0548	.581	.704	117.81
1976	159.50	1143.43	.139	.327	.0692	.623	.800	130.30
1977	172.29	1244.44	.138	.335	.0724	n.a.	n.a.	152.08
1978	184.58	1386.22	.133	.328	.0706	n.a.	n.a.	178.48

NOTE: See the text for definitions and data sources.

5.3 Equation (30). Aggregate investment regression

Read.

- Estimated slope on modified q : 0.0423.
- Intercept: 0.0980.
- $R^2 = 0.46$.
- Durbin–Watson: 0.43.

Economic message. Modified q is positively related to aggregate investment, which is consistent with the model. But the residual time-series dependence is substantial, so the regression should be read as a rough empirical illustration rather than a decisive econometric test.

6 Short summary

- Hayashi shows that Tobin’s q -theory is the neoclassical investment model with adjustment costs written in shadow-price form.
- The firm’s optimal investment rate depends on *modified* marginal q , not raw average q .
- Average q can proxy for marginal q only under a competitive constant-returns benchmark, and even then tax adjustments matter.
- If the firm has market power, average q exceeds marginal q because it includes monopoly rent.
- In the aggregate U.S. data, tax-adjusted modified q helps explain investment, but the empirical evidence is illustrative rather than definitive.

7 Conclusion

Core conclusion. Hayashi's paper is the classic bridge between investment theory and investment empirics. It explains exactly why a properly defined marginal q summarizes the firm's investment incentives and when observable average q can stand in for it.

Economic intuition. Adjustment costs make installed capital scarce at the margin. The shadow value of that scarce installed capital is marginal q . If firms are competitive and technologies scale proportionally, the marginal value of capital equals its average value. Once taxes or monopoly rents enter, that simple equality fails unless one corrects for them explicitly.

Takeaway for empirical work. Regressing investment on average q is not innocuous. It is justified only under specific assumptions, and the correct empirical object is usually a tax-adjusted modified q , not the raw market-value ratio.